

STAT 331 Applied Linear Models

University of Waterloo - Fall 2025

11th September 2025

Instructor: Kun Liang

L^AT_EX: Xing Liu

Contents

1	Simple Linear Regression (SLR)	2
1.1	Introduction	2
1.2	Least Squares Estimation	3

1 Simple Linear Regression (SLR)

Lecture 2, 2025/09/09

1.1 Introduction

Definition 1.1 (SLR Population Model).

Population model.

$$\text{Response: } y = \beta_0 + \beta_1 x + \epsilon,$$

where β_0, β_1 are regression coefficients, x is the predictor, and ϵ is the random error.

Definition 1.2 (SLR Sample Model).

Sample: n pairs of observations $(x_i, y_i), i = 1, 2, \dots, n$.

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i,$$

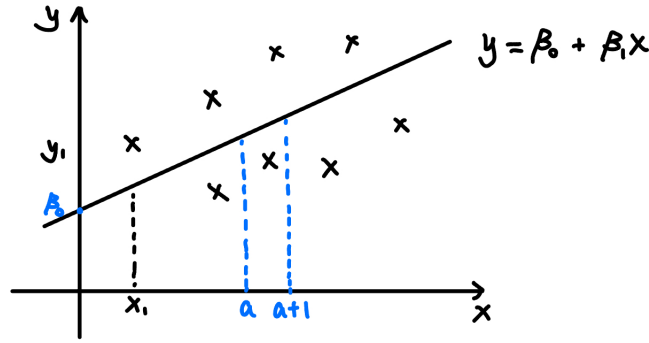
where β_0 is the intercept and β_1 is the slope.

	Fixed/Random	Known/Unknown
x_i 's	Fixed	Known
β_i 's	Fixed	Known
ϵ_i 's	Random	Unknown
y_i 's	Random	Known

SLR Assumptions

- (1) $\mathbb{E}[\epsilon_i] = 0 \implies \mathbb{E}[y_i] = \beta_0 + \beta_1 x_i$.
- (2) $\epsilon_1, \dots, \epsilon_n$ are statistically independent.
- (3) $\text{Var}(\epsilon_i) = \sigma^2 \implies \text{Var}(y_i) = \sigma^2$.
- (4) ϵ_i is normally distributed, i.e. $\epsilon_i \sim N(0, \sigma^2) \implies y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$.

Note. The first three assumptions are called the **Gauss-Markov assumptions**.



The slope β_1 is often of primary interest. We have the following interpretations of β_1 .

- (1) $\beta_1 = \mathbb{E}[y \mid x = a + 1] - \mathbb{E}[y \mid x = a]$.
- (2) If $\beta_1 = 0$, then $\mathbb{E}[y \mid x] = \beta_0$, i.e. y does not depend on x .

1.2 Least Squares Estimation

Task: Given the sample observation (x_i, y_i) , $i = 1, 2, \dots, n$, we want to find (β_0, β_1) such that

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

is minimized.

To minimize the objective function $S(\beta_0, \beta_1)$, we set the partial derivatives to zero:

$$\frac{\partial S(\beta_0, \beta_1)}{\partial \beta_0} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0, \quad (1.1)$$

$$\frac{\partial S(\beta_0, \beta_1)}{\partial \beta_1} = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) x_i = 0. \quad (1.2)$$

Then, to solve for β_0, β_1 ,

$$(1.1) \implies n\beta_0 = \sum_{i=1}^n y_i - \beta_1 \sum_{i=1}^n x_i, \text{ so}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}, \text{ where } \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \text{ and } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

(2.1) $\implies \sum_{i=1}^n x_i y_i - \beta_0 \sum_{i=1}^n x_i - \beta_1 \sum_{i=1}^n x_i^2 = 0$. Substitute β_0 into this, we have

$$\begin{aligned} \sum_{i=1}^n x_i y_i - (\bar{y} - \beta_1 \bar{x}) \sum_{i=1}^n x_i - \beta_1 \sum_{i=1}^n x_i^2 &= 0 \\ \sum_{i=1}^n x_i y_i - \bar{y} \sum_{i=1}^n x_i + \beta_1 \bar{x} \sum_{i=1}^n x_i - \beta_1 \sum_{i=1}^n x_i^2 &= 0 \\ \beta_1 &= \frac{\sum_{i=1}^n x_i y_i - n \bar{x} \bar{y}}{\sum_{i=1}^n x_i^2 - n \bar{x}^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}, \end{aligned}$$

Lecture 3, 2025/09/11

Recall from STAT 231:

$$\text{Sample variance: } S_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2,$$

$$\text{Sample covariance: } S_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

Then,

$$\beta_1 = \frac{S_{xy}}{S_x^2}.$$

Definition 1.3 (Pearson Correlation Coefficient).

The **Pearson correlation coefficient** between x and y is

$$\rho_{xy} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}.$$

Definition 1.4 (Sample Correlation Coefficient).

The **sample correlation coefficient** between x and y is

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}.$$

Then, we have $\beta_1 = r_{xy} \frac{\sqrt{S_{yy}}}{\sqrt{S_{xx}}}$.

Thus, the least squares estimates (LSEs) are:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \quad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}.$$

Proposition 1.1 (Properties of $\hat{\beta}_0$ and $\hat{\beta}_1$).

- (1) The least squares estimates (LSEs) are unbiased, i.e. $\mathbb{E}[\hat{\beta}_0] = \beta_0, \mathbb{E}[\hat{\beta}_1] = \beta_1$.
- (2) $\text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right), \text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}$.
- (3) $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{x}}{S_{xx}}$.

Proof.

(1) We have

$$\begin{aligned} \mathbb{E}[\hat{\beta}_1] &= \mathbb{E}\left[\frac{S_{xy}}{S_{xx}}\right] = \mathbb{E}\left[\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{S_{xx}}\right] \\ &= \mathbb{E}\left[\frac{\sum_{i=1}^n (x_i - \bar{x})y_i}{S_{xx}}\right] \end{aligned}$$

Note that $\hat{\beta}_1 = \sum_{i=1}^n c_i y_i$, where $c_i = \frac{x_i - \bar{x}}{S_{xx}}$.

Remark (Properties of c_i 's).

- (i) $\sum_{i=1}^n c_i = 0$.
- (ii) $\sum_{i=1}^n c_i x_i = \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x})x_i = \frac{1}{S_{xx}} (\sum_{i=1}^n x_i^2 - n\bar{x}^2) = 1$.
- (iii) $\sum_{i=1}^n c_i^2 = \frac{1}{S_{xx}^2} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{S_{xx}}$.

Then,

$$\begin{aligned} \mathbb{E}[\hat{\beta}_1] &= \mathbb{E}\left[\sum_{i=1}^n c_i y_i\right] = \sum_{i=1}^n c_i \mathbb{E}[y_i] \\ &= \sum_{i=1}^n c_i (\beta_0 + \beta_1 x_i) = \beta_0 \sum_{i=1}^n c_i + \beta_1 \sum_{i=1}^n c_i x_i = \beta_1. \end{aligned}$$

Similarly,

$$\begin{aligned}
\mathbb{E}[\hat{\beta}_0] &= \mathbb{E}[\bar{y} - \hat{\beta}_1 \bar{x}] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[y_i] - \bar{x} \mathbb{E}[\hat{\beta}_1] \\
&= \frac{1}{n} \sum_{i=1}^n (\beta_0 + \beta_1 x_i) - \bar{x} \beta_1 \\
&= \beta_0 + \beta_1 \bar{x} - \bar{x} \beta_1 = \beta_0.
\end{aligned}$$

(2) We have

$$\begin{aligned}
\text{Var}(\hat{\beta}_1) &= \text{Var}\left(\sum_{i=1}^n c_i y_i\right) = \sum_{i=1}^n c_i^2 \text{Var}(y_i) \\
&= \sigma^2 \sum_{i=1}^n c_i^2 = \frac{\sigma^2}{S_{xx}}.
\end{aligned}$$

Similarly,

$$\begin{aligned}
\text{Var}(\hat{\beta}_0) &= \text{Var}(\bar{y} - \hat{\beta}_1 \bar{x}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n y_i - \bar{x} \sum_{i=1}^n c_i y_i\right) \\
&= \text{Var}\left(\sum_{i=1}^n \left(\frac{1}{n} - c_i \bar{x}\right) y_i\right) = \sum_{i=1}^n \left(\frac{1}{n} - c_i \bar{x}\right)^2 \text{Var}(y_i) \\
&= \sigma^2 \sum_{i=1}^n \left(\frac{1}{n} - c_i \bar{x}\right)^2 = \sigma^2 \left(\sum_{i=1}^n \frac{1}{n^2} - 2 \frac{\bar{x}}{n} \sum_{i=1}^n c_i + \bar{x}^2 \sum_{i=1}^n c_i^2\right) \\
&= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right).
\end{aligned}$$

(3) We have

$$\begin{aligned}
\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) &= \text{Cov}(\bar{y} - \hat{\beta}_1 \bar{x}, \hat{\beta}_1) \\
&= \text{Cov}\left(\sum_{i=1}^n \left(\frac{1}{n} - c_i \bar{x}\right) y_i, \sum_{i=1}^n c_i y_i\right) \\
&= \sum_{i=1}^n \sum_{j=1}^n \left(\frac{1}{n} - c_i \bar{x}\right) c_j \text{Cov}(y_i, y_j)
\end{aligned}$$

Note that $\text{Cov}(y_i, y_j) = \begin{cases} \sigma^2, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$ Then,

$$\begin{aligned} \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) &= \sigma^2 \sum_{i=1}^n \left(\frac{1}{n} - c_i \bar{x} \right) c_i \\ &= \sigma^2 \left(\frac{1}{n} \sum_{i=1}^n c_i - \bar{x} \sum_{i=1}^n c_i^2 \right) = -\frac{\sigma^2 \bar{x}}{S_{xx}}. \end{aligned} \quad \square$$

Definition 1.5 (Residual).

The **residual** for the i -th observation is the difference between the observed and the fitted value:

$$r_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i.$$

Under the least squares fit, we have the following.

Proposition 1.2 (Properties of Residuals).

- (1) $\sum_{i=1}^n r_i = 0.$
- (2) $\sum_{i=1}^n r_i x_i = 0.$
- (3) $\sum_{i=1}^n r_i \hat{y}_i = 0$, where $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i.$